

The background is a solid dark blue color. It is decorated with numerous gold-colored splatters and dots of varying sizes. These splatters are concentrated in the top right and bottom left corners, with a few scattered dots throughout the rest of the page.

ADVANCED DOM MANIPULATION (EVENT DELEGATION, BUBBLING, CAPTURING)

I. Event Bubbling (Bottom → Up)

When an event occurs on a child element, it "bubbles up" to its parent and then to the entire document.

Example: Event Bubbling in Action

```
<div id="parent">
  <button id="child">Click Me</button>
</div>

<script>

document.getElementById("child").addEventListener("click
", function () {
  console.log("Child Clicked!");
});
```

```
document.getElementById("parent").addEventListener("click", function () {  
    console.log("Parent Clicked!");  
});  
</script>
```

◆ Clicking the button logs:

1 "Child Clicked!"

2 "Parent Clicked!" (bubbles up to parent)

2. Event Capturing (Top → Down) ▼

Event Capturing (or Trickle Down) is the opposite of bubbling – the event fires first on the topmost element and moves down to the target.

✓ Example: Event Capturing

```
<div id="parent">
  <button id="child">Click Me</button>
</div>

<script>
  document.getElementById("parent").addEventListener(
    "click",
    function () {
      console.log("Parent Clicked!");
    },
    true // Enables event capturing
  );

  document.getElementById("child").addEventListener("click"
, function () {
    console.log("Child Clicked!");
  });
</script>
```

◆ Clicking the button logs:

1 "Parent Clicked!" (captures event first)

2 "Child Clicked!"

3. Event Delegation (Efficient Event Handling)

Instead of adding event listeners to multiple elements, attach one listener to a common ancestor and use `event.target` to identify the clicked element.

✓ Example: Handling Multiple Buttons with One Event Listener

```
<ul id="list">
  <li>Item 1</li>
  <li>Item 2</li>
  <li>Item 3</li>
</ul>

<script>

document.getElementById("list").addEventListener
("click", function (event) {
  if (event.target.tagName === "LI") {
    console.log("You clicked:",
event.target.innerText);
  }
});
</script>
```

- ◆ Clicking any logs "You clicked: Item X".
- ◆ No need to attach listeners to each list item

4. Stopping Event Propagation

Prevent Bubbling with `stopPropagation()`

```
document.getElementById("child").addEventListener("click", function (event) {  
    event.stopPropagation(); // Stops bubbling  
    console.log("Child Clicked!");  
});
```

- ◆ Now, clicking the button won't trigger the parent's event.